

GED Science Practice Test 12

PART A — QUESTIONS

GED-SCI12-001

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies how evening light exposure affects melatonin, a hormone involved in sleep timing. Two groups of adults follow the same sleep schedule for one week. Group A uses dim room light for one hour before bed. Group B uses bright overhead light for one hour before bed. Melatonin level is measured at bedtime.

Group | Evening light condition | Average melatonin level (pg/mL)

A | Dim light | 48

B | Bright light | 26

Which conclusion is best supported by the data?

- A) Bright evening light was associated with lower bedtime melatonin levels.
- B) Bright evening light increased bedtime melatonin levels.
- C) Light exposure had no relationship to melatonin level.
- D) Melatonin levels were identical in both groups.

GED-SCI12-002

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

Impulse can be calculated as $\text{force} \times \text{time}$.

Collision | Average force (N) | Contact time (s)

A | 80 | 0.5

B | 120 | 0.4

C | 60 | 1.0

D | 200 | 0.2

Which collision has the greatest impulse?

- A) Collision A
- B) Collision B
- C) Collision C
- D) Collision D

GED-SCI12-003

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A geologist studies pumice, a volcanic rock that contains many holes. These holes formed when gas bubbles were trapped in lava as it cooled quickly during an eruption. The rock is lightweight and has a glassy texture.

Which statement best describes how pumice forms?

- A) Lava cools quickly while gas bubbles are trapped inside.
- B) Sand grains are compacted at the bottom of a river.
- C) Existing rock is changed by heat and pressure without melting.
- D) Metal in Earth's core solidifies into crystals.

GED-SCI12-004

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Ammonia is a toxic waste product formed when the body breaks down proteins. The liver converts ammonia into urea, a less toxic substance. Urea then travels in the blood to the kidneys, where it can be removed in urine.

Which statement best describes the role of the liver in this process?

- A) The liver changes a toxic waste into a less toxic substance.
- B) The liver pumps oxygen-rich blood to the lungs.
- C) The liver filters air before it enters the nose.
- D) The liver stores genetic information in chromosomes.

GED-SCI12-005

Section: Science

Type: numeric_entry

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A person pushes a box with a force of 75 N for 4 meters.

Work = force $\times$ distance

How much work is done on the box in joules?

Type your answer as a whole number.

GED-SCI12-006

Section: Science

Type: multi_select

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A student tests how the roughness of a surface affects the force needed to start moving a block. The same block is placed on four surfaces with different roughness levels. The student uses the same spring scale in every trial.

Which TWO factors should be kept constant?

- A) Mass of the block
- B) Spring scale used
- C) Roughness of the surface
- D) Force needed to start motion
- E) Final measured value

GED-SCI12-007

Section: Science

Type: dropdown

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A trait is controlled by several genes, and environmental factors such as nutrition can also affect the trait. Human height is an example of [D1] inheritance because it is influenced by [D2].

Dropdowns:

D1) polygenic | single-celled | sedimentary

D2) multiple genes | only one mineral | ocean tides

GED-SCI12-008

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student uses a convex lens to focus sunlight onto a small spot on a card. The lens bends incoming light rays so they meet at a focal point.

Which property of light is mainly being used?

- A) Refraction
- B) Fermentation
- C) Blood clotting
- D) Rock compaction

GED-SCI12-009

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A region depends heavily on groundwater for farming. Over several years, water is pumped from underground faster than it is naturally replaced. Wells must be drilled deeper, and some shallow wells stop producing water.

Which conclusion is best supported?

- A) Groundwater is being withdrawn faster than it is replenished.
- B) Groundwater cannot be affected by human use.
- C) Pumping water creates new aquifers instantly.
- D) Deeper wells prove that groundwater levels are rising.

GED-SCI12-010

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A clinic measures cortisol, a stress hormone, in volunteers before and after a relaxation program.

Group | Average cortisol before ($\mu\text{g/dL}$) | Average cortisol after ($\mu\text{g/dL}$)

Relaxation program | 18 | 12

No program | 17 | 16

Which claim is best supported by the data?

- A) The relaxation group showed a larger decrease in cortisol.
- B) The no-program group showed a larger decrease in cortisol.
- C) Cortisol increased in both groups.
- D) Cortisol stayed exactly the same in both groups.

GED-SCI12-011

Section: Science

Type: drag_drop

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Match each target with the correct tile.

Tiles:

- A) Shape of the wing model
- B) Lift force measured by the sensor
- C) Same fan speed used for every trial
- D) Color of the data table

Targets:

- T1) Independent variable
- T2) Dependent variable
- T3) Controlled variable

GED-SCI12-012

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

Some neurons are covered by myelin, a fatty layer that helps electrical impulses travel faster along the axon. When myelin is damaged, nerve signals may slow down or fail to reach their target properly.

Which statement best describes the role of myelin?

- A) It helps nerve impulses travel efficiently along neurons.
- B) It digests food in the small intestine.
- C) It stores minerals in volcanic rock.
- D) It changes carbon dioxide into oxygen in the lungs.

GED-SCI12-013

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A pendulum swings back and forth. At the highest point of the swing, the pendulum has more gravitational potential energy and less kinetic energy. At the lowest point, it moves fastest and has more kinetic energy.

Which statement best describes the energy change?

- A) Energy shifts between gravitational potential energy and kinetic energy.
- B) Energy is created only at the lowest point.
- C) Energy disappears at the highest point.
- D) Kinetic energy changes into DNA.

GED-SCI12-014

Section: Science

Type: numeric_entry

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A solar farm produces 18,000 kilowatt-hours of energy in 6 hours.

What is the average energy production rate in kilowatt-hours per hour?

Type your answer as a whole number.

GED-SCI12-015

Section: Science

Type: mcq

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A researcher studies whether a high-fiber breakfast affects how full people feel before lunch. One group eats a high-fiber breakfast, and another group eats a low-fiber breakfast with the same number of calories. Both groups rate fullness after three hours.

Why should the breakfasts have the same number of calories?

- A) To help isolate fiber amount as the main variable being tested
- B) To make fiber and calories the same thing
- C) To prevent people from rating fullness
- D) To guarantee that both groups give identical answers

GED-SCI12-016

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student compares the strength of different bridge shapes made from the same number of craft

sticks. Each bridge is placed between the same supports, and weights are added until the bridge bends 2 cm.

What is the dependent variable?

- A) Amount of weight needed to bend the bridge 2 cm
- B) Shape of the bridge
- C) Number of craft sticks used
- D) Distance between the supports

GED-SCI12-017

Section: Science

Type: dropdown

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A first-class lever has the fulcrum between the input force and the load. Moving the fulcrum closer to the load can [D1] the input force needed, but the input side must move through a [D2] distance.

Dropdowns:

D1) decrease | genetically copy | evaporate

D2) greater | smaller | zero-only

GED-SCI12-018

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A doctor measures urine concentration in a patient before and after the release of antidiuretic hormone (ADH). ADH causes the kidneys to reabsorb more water.

Condition | Urine volume in 4 hours (mL) | Urine concentration

Before ADH increase | 620 | Low

After ADH increase | 280 | High

Which statement is best supported by the data?

- A) After ADH increased, urine volume decreased and urine became more concentrated.
- B) After ADH increased, urine volume increased and urine became less concentrated.
- C) ADH caused the kidneys to stop interacting with water.
- D) Urine volume and concentration stayed the same.

GED-SCI12-019

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records the distance a braking cart travels after different brake-pad materials are used. The cart starts at the same speed in every trial.

Brake-pad material | Stopping distance (m)

Material A | 3.8

Material B | 2.9
Material C | 4.5
Material D | 2.1

Which material produced the shortest stopping distance?

- A) Material A
- B) Material B
- C) Material C
- D) Material D

GED-SCI12-020

Section: Science

Type: multi_select

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

Select TWO answers.

A state tracks electricity generated from two resources.

Year	Coal electricity (million MWh)	Wind electricity (million MWh)
1	92	18
2	81	27
3	70	39
4	61	52

Which TWO statements are best supported by the data?

- A) Coal electricity decreased from Year 1 to Year 4.
- B) Wind electricity increased from Year 1 to Year 4.
- C) Coal electricity increased every year.
- D) Wind electricity was highest in Year 1.
- E) Both resources produced exactly the same amount every year.

GED-SCI12-021

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A vaccine causes the immune system to produce memory cells. Years later, if the same pathogen enters the body, memory cells help white blood cells respond more quickly and produce antibodies faster.

Which statement best describes the role of memory cells?

- A) They help the immune system respond faster to a later exposure.
- B) They pump blood through the heart.
- C) They form lava during volcanic eruptions.
- D) They prevent cells from making proteins.

GED-SCI12-022

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student places a book on a desk. Gravity pulls the book downward, and the desk pushes upward with an equal force. The book remains still because the forces cancel.

Which statement best describes the forces?

- A) The forces are balanced.
- B) The downward force is the only force.
- C) The book must accelerate upward.
- D) The desk force removes the book's mass.

GED-SCI12-023

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Reading for Meaning in Science

Prompt:

A text model describes how large mountain ranges can form. Two continental plates slowly move toward each other. Because both plates are relatively thick and buoyant, neither plate easily sinks. The crust folds, thickens, and rises over millions of years.

Which landform is most likely produced by this process?

- A) Folded mountain range
- B) Ocean trench made by oceanic subduction
- C) River delta formed from deposited sediment
- D) Crater formed by a meteor impact

GED-SCI12-024

Section: Science

Type: numeric_entry

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A study follows 360 adults. Of these adults, 54 have a family history of a certain inherited disorder.

What percent of the adults have a family history of the disorder?

Type your answer as a percent rounded to the nearest whole number.

GED-SCI12-025

Section: Science

Type: mcq

Category: Physical Science

Skill: Designing and Interpreting Science Experiments

Prompt:

A student tests how the angle of a solar panel affects energy output. The panel is placed at four different angles on the same sunny day. The same panel and same measuring device are used each time.

Which factor is the independent variable?

- A) Angle of the solar panel
- B) Energy output
- C) Same solar panel
- D) Same measuring device

GED-SCI12-026

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

The rough endoplasmic reticulum has ribosomes attached to its surface. Proteins made by these ribosomes may enter the rough endoplasmic reticulum, where they begin to fold and be prepared for transport.

Which statement best describes the rough endoplasmic reticulum?

- A) It helps process proteins made by attached ribosomes.
- B) It stores most groundwater in aquifers.
- C) It changes oxygen into rock crystals.
- D) It produces earthquakes at plate boundaries.

GED-SCI12-027

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student records the output of a hand-crank generator.

Crank turns per minute: 20, 40, 60, 80

Energy produced in 1 min (J): 15, 31, 45, 62

Which statement is best supported by the data?

- A) More crank turns per minute were associated with more energy produced.
- B) More crank turns per minute were associated with less energy produced.
- C) Energy produced stayed exactly the same at all crank speeds.
- D) Cranking had no relationship to energy output.

GED-SCI12-028

Section: Science

Type: mcq

Category: Earth and Space Science

Skill: Using Numbers and Graphics in Science

Prompt:

A student compares four rocky objects in space.

Object | Diameter (km)

A | 12

B | 48

C | 520
D | 1,200

Which object has the largest diameter?

- A) Object A
- B) Object B
- C) Object C
- D) Object D

GED-SCI12-029

Section: Science

Type: mcq

Category: Life Science

Skill: Using Numbers and Graphics in Science

Prompt:

A neurologist compares average nerve signal speed in two groups.

Group | Average nerve signal speed (m/s)

Healthy myelin | 85

Damaged myelin | 34

Which claim is best supported by the data?

- A) Damaged myelin was associated with slower nerve signal speed.
- B) Damaged myelin was associated with faster nerve signal speed.
- C) Myelin condition had no relationship to nerve signal speed.
- D) Both groups had exactly the same nerve signal speed.

GED-SCI12-030

Section: Science

Type: mcq

Category: Physical Science

Skill: Reading for Meaning in Science

Prompt:

A student mixes an acidic cleaner with a basic cleaner and notices that the mixture becomes warmer. The labels warn not to mix the cleaners because the reaction can release heat and produce harmful fumes.

Which statement best describes the heat change?

- A) The reaction is exothermic because thermal energy is released.
- B) The reaction is endothermic because thermal energy is released.
- C) The reaction is not chemical because the mixture becomes warm.
- D) The reaction stops all energy transfer.

GED-SCI12-031

Section: Science

Type: multi_select

Category: Life Science

Skill: Designing and Interpreting Science Experiments

Prompt:

Select TWO answers.

A researcher studies whether a new balance board improves ankle stability. Participants use the

board for different practice times each day. Ankle stability is measured with the same balance test after four weeks.

Which TWO steps would improve the investigation?

- A) Use the same balance test for all participants.
- B) Include multiple participants at each practice-time level.
- C) Change the balance test for each participant.
- D) Record only participants with the best scores.
- E) Use different study lengths for different participants.

GED-SCI12-032

Section: Science

Type: mcq

Category: Physical Science

Skill: Using Numbers and Graphics in Science

Prompt:

A machine does 360 J of work in 12 seconds.

Power = work $\div$ time

What is the machine's power?

- A) 12 W
- B) 30 W
- C) 360 W
- D) 4,320 W

GED-SCI12-033

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A forest is damaged by a windstorm that knocks down many trees but leaves soil and some roots in place. In the following years, grasses and shrubs grow quickly, and tree seedlings begin to appear.

Which ecological process is most directly described?

- A) Secondary succession
- B) Lunar eclipse
- C) Chemical neutralization
- D) Mechanical advantage

GED-SCI12-034

Section: Science

Type: mcq

Category: Life Science

Skill: Reading for Meaning in Science

Prompt:

A person inherits one changed copy of a gene involved in DNA repair. The changed copy does not guarantee disease, but it can increase the person's risk for certain cancers because damaged DNA may be repaired less effectively.

Which statement is best supported?

- A) Some inherited gene changes can increase disease risk without guaranteeing disease.
- B) Inherited gene changes never affect health.
- C) DNA repair genes are found only in bacteria.
- D) A changed gene always prevents all cell division.

PART B — ANSWER KEY + EXPLANATIONS

GED-SCI12-001 — Correct: A — Correct Answer: Bright evening light was associated with lower bedtime melatonin levels. — Explanation: The bright-light group had a lower average melatonin level than the dim-light group.

GED-SCI12-002 — Correct: C — Correct Answer: Collision C — Explanation: Impulse = force $\times$ time. Collision C has $60 \text{ N} \times 1.0 \text{ s} = 60 \text{ N}\cdot\text{s}$, the greatest value.

GED-SCI12-003 — Correct: A — Correct Answer: Lava cools quickly while gas bubbles are trapped inside. — Explanation: Pumice forms when gas bubbles are trapped in lava that cools quickly.

GED-SCI12-004 — Correct: A — Correct Answer: The liver changes a toxic waste into a less toxic substance. — Explanation: The liver converts toxic ammonia into less toxic urea.

GED-SCI12-005 — Correct: 300 — Tolerance: exact — Correct Answer: 300 — Explanation: Work = $75 \text{ N} \times 4 \text{ m} = 300 \text{ J}$.

GED-SCI12-006 — Correct: A,B — Correct Answer: Mass of the block | Spring scale used — Explanation: These factors should stay constant so surface roughness is the main variable being tested.

GED-SCI12-007 — Correct: D1=polygenic; D2=multiple genes — Correct Answer: D1=polygenic; D2=multiple genes — Explanation: Polygenic traits are influenced by more than one gene.

GED-SCI12-008 — Correct: A — Correct Answer: Refraction — Explanation: A convex lens focuses light by bending, or refracting, light rays.

GED-SCI12-009 — Correct: A — Correct Answer: Groundwater is being withdrawn faster than it is replenished. — Explanation: Falling well levels and deeper drilling indicate groundwater is being used faster than it is replaced.

GED-SCI12-010 — Correct: A — Correct Answer: The relaxation group showed a larger decrease in cortisol. — Explanation: The relaxation group decreased by $6 \mu\text{g/dL}$, compared with $1 \mu\text{g/dL}$ in the no-program group.

GED-SCI12-011 — Correct: T1=A; T2=B; T3=C — Correct Answer: T1=Shape of the wing model; T2=Lift force measured by the sensor; T3=Same fan speed used for every trial — Explanation: Wing shape is changed, lift force is measured, and fan speed is controlled.

GED-SCI12-012 — Correct: A — Correct Answer: It helps nerve impulses travel efficiently along neurons. — Explanation: Myelin helps nerve impulses move quickly along axons.

GED-SCI12-013 — Correct: A — Correct Answer: Energy shifts between gravitational potential energy and kinetic energy. — Explanation: A pendulum changes energy form as height and speed change during the swing.

GED-SCI12-014 — Correct: 3000 — Tolerance: exact — Correct Answer: 3000 — Explanation: $18,000 \text{ kilowatt-hours} \div 6 \text{ hours} = 3,000 \text{ kilowatt-hours per hour}$.

GED-SCI12-015 — Correct: A — Correct Answer: To help isolate fiber amount as the main variable being tested — Explanation: Keeping calories the same helps test whether fiber amount affects fullness.

GED-SCI12-016 — Correct: A — Correct Answer: Amount of weight needed to bend the bridge 2 cm — Explanation: The dependent variable is what is measured after changing bridge shape.

GED-SCI12-017 — Correct: D1=decrease; D2=greater — Correct Answer: D1=decrease; D2=greater — Explanation: Moving the fulcrum closer to the load can reduce input force, but the input side must move farther.

GED-SCI12-018 — Correct: A — Correct Answer: After ADH increased, urine volume decreased and urine became more concentrated. — Explanation: ADH causes more water reabsorption, resulting in lower urine volume and higher concentration.

GED-SCI12-019 — Correct: D — Correct Answer: Material D — Explanation: Material D produced the shortest stopping distance, 2.1 m.

GED-SCI12-020 — Correct: A,B — Correct Answer: Coal electricity decreased from Year 1 to Year 4. | Wind electricity increased from Year 1 to Year 4. — Explanation: Coal decreased from 92 to 61 million MWh, while wind increased from 18 to 52 million MWh.

GED-SCI12-021 — Correct: A — Correct Answer: They help the immune system respond faster to a later exposure. — Explanation: Memory cells help the immune system respond more quickly if the same pathogen appears again.

GED-SCI12-022 — Correct: A — Correct Answer: The forces are balanced. — Explanation: Equal upward and downward forces cancel, so the book remains still.

GED-SCI12-023 — Correct: A — Correct Answer: Folded mountain range — Explanation: Continental collision can fold and thicken crust, forming mountain ranges.

GED-SCI12-024 — Correct: 15 — Tolerance: exact — Correct Answer: 15 — Explanation: $54 \div 360 \times 100 = 15\%$.

GED-SCI12-025 — Correct: A — Correct Answer: Angle of the solar panel — Explanation: The independent variable is the factor intentionally changed by the student.

GED-SCI12-026 — Correct: A — Correct Answer: It helps process proteins made by attached ribosomes. — Explanation: Rough endoplasmic reticulum helps fold and process proteins made by ribosomes.

GED-SCI12-027 — Correct: A — Correct Answer: More crank turns per minute were associated with more energy produced. — Explanation: Energy output increased as crank turns per minute increased.

GED-SCI12-028 — Correct: D — Correct Answer: Object D — Explanation: Object D has the largest listed diameter, 1,200 km.

GED-SCI12-029 — Correct: A — Correct Answer: Damaged myelin was associated with slower nerve signal speed. — Explanation: The damaged-myelin group had a lower average signal speed than the healthy-myelin group.

GED-SCI12-030 — Correct: A — Correct Answer: The reaction is exothermic because thermal energy is released. — Explanation: A reaction that releases thermal energy is exothermic.

GED-SCI12-031 — Correct: A,B — Correct Answer: Use the same balance test for all participants. | Include multiple participants at each practice-time level. — Explanation: A consistent test and repeated participants at each level improve reliability.

GED-SCI12-032 — Correct: B — Correct Answer: 30 W — Explanation: $\text{Power} = 360 \text{ J} \div 12 \text{ s} = 30 \text{ W}$.

GED-SCI12-033 — Correct: A — Correct Answer: Secondary succession — Explanation: Secondary succession occurs after a disturbance when soil remains.

GED-SCI12-034 — Correct: A — Correct Answer: Some inherited gene changes can increase disease risk without guaranteeing disease. — Explanation: Some mutations increase risk by affecting repair processes, but they do not guarantee a disease will occur.